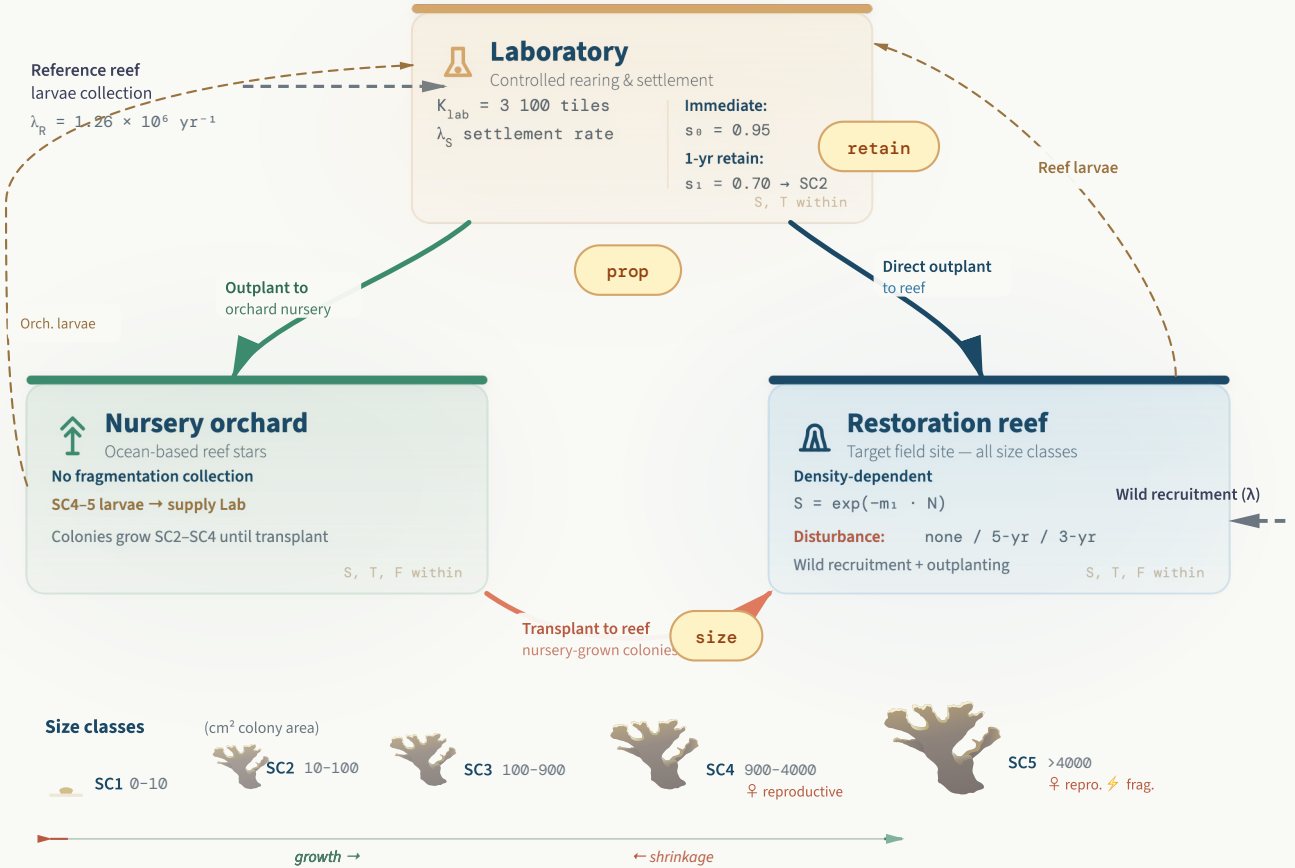
Laboratory
Controlled rearing & settlement
K_{lab} = 3 100 tiles
λ_S settlement rate
Immediate:
s_0 = 0.95
1-yr retain:
s_1 = 0.70 → SC2
S, T within
retain
Reef larvae
Reference reef
larvae collection
λ_R = 1.26 × 10^6 yr^{-1}
prop
Outplant to
orchard nursery
Direct outplant
to reef
Orch. larvae
Nursery orchard
Ocean-based reef stars
No fragmentation collection
SC4–5 larvae → supply Lab
Colonies grow SC2–SC4 until transplant
S, T, F within
Restoration reef
Target field site — all size classes
Density-dependent
S = exp(−m_1 · N)
Disturbance: none / 5-yr / 3-yr
Wild recruitment + outplanting
S, T, F within
Wild recruitment (λ)
Transplant to reef
nursery-grown colonies
size
Size classes
(cm² colony area)
SC1 0–10
SC2 10–100
SC3 100–900
SC4 900–4000
♀ reproductive
SC5 >4000
♀ repro. ⚡ frag.
growth →
← shrinkage
N(t+1) = (T + F) · (S ⊙ N(t)) + R
Outplant
Transplant
Direct outplant
External input
Larvae feedback
decision
Decision point
S survival
T transition
F fragmentation
R recruitment
⊙ element-wise
Annual time step · 50 yr · 100 replicates